# Simple Simulation I: School Closure

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# 1 Experiment Overview

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## 1.1 Aim

The aim of the experiment is observing LLM-driven deliberation between an MOE Official and an affected School Teacher on Singapore school closures as a result of mergers.

> *Due to declining birth rates, the Ministry of Education (MOE) regularly merges or closes primary and secondary schools. This is a highly emotional topic, pitting practical resource allocation and diverse social mixing against heritage, alumni loyalty, and localized community disruption.*

Reference: CNA, 2021.

In this experiment, the MOE Official is strongly supportive of the school merger, while the School Teacher is strongly against their school being closed.

## 1.2 Setup

### 1.2.1 Hypotheses

$\boldsymbol{H}_0$: Requiring an internal reflection step for an agent before they can respond in a deliberation does NOT significantly affect:

1. The semantic convergence of the deliberation.
2. Both agents' willingness to change their stance.
3. The discourse quality of the deliberation.

$\boldsymbol{H}_1$: Requiring an internal reflection step IMPROVES the above metrics.

### 1.2.2 Tests

**Control Run**: Both agents read the chat history and directly output their next dialogue response.

**Treatment Run**: Both agents are prompted to generate a `private_scratchpad` (internal reflection) evaluating the arguments made by the other before outputting their `public_response`.

## 1.3 Indicators

### 1.3.1 Variables

**Independent variable**: The presence of an internal scratchpad.

**Dependent variables**:

1. **Convergence** Cosine similarity between the final stances of the different agents. [Semantic convergence]
2. **Self-Shift**: Cosine distance between each agent's original and final stances, averaged across all agents. [Agents' willingness to change their stance]
3. **Plurality/CovD**: The determinant of the covariance matrix of the stances. [Semantic convergence]
4. **DQI**: The Discourse Quality Index. [Discourse Quality]

## 1.4 Interpretation

1. **Convergence**: The larger the convergence metric, the stronger the deliberation consensus.
2. **Self-Shift**: The larger the self-shift metric, the more all agents' willingness to change their stances.
3. **Plurality/CovD**: The higher the CovD, the higher the diversity of opinions (plurality) and the weaker the consensus.
4. **DQI**: The higher the DQI, the better the discourse quality.

## 1.5 Execution Flow

1. User prepares a `data/config.json` and loads-in an API key and/or Base URL in `.env`.
2. User executes `src/main.py --config "path_to_config.json"`.
3. Script programmatically extracts config details and initializes client for compute service.
4. Agents generate pre-deliberation stances[1].
5. For each run (Control/Treatment), agents take turns responding to one another for a config-specified $N$ rounds.
6. Agents generate a final 1-paragraph post-deliberation stance after their turns are fully concluded.
7. Raw JSON chat/prompt transcripts are saved.
8. Raw JSON metrics are saved.

In essence, the contract of the repo is:

> **IF** the user specifies the `data/config.json` properly and loads in a valid API key and/or Base URL in `.env`,
>
> **THEN** the pipeline will run an $N$-turn deliberation between 2 agents using the specified model(s) for both the Control & Treatment, and output the chat histories of each run in separate `JSON` files. The pipeline will also output an `evals.json` with the metrics: Convergence, Self-Shift, Plurality/CovD, and DQI. All outputs will be saved in an automatically created & indexed sub-directory of `data`.

A sample `results_x.json` looks like:

```
{
  "stances_pre": {
    "MOE Official": "...",
    "Teacher": "..."
  },
  "stances_post": {
    "MOE Official": "...",
    "Teacher": "..."
  },
  "chat_history": {
    "turn_1": [
      {
        "agent": "MOE Official",
        "public_response": "...",
        "private_scratchpad": "...",
      },
      {
        "agent": "Teacher",
        "public_response": "...",
        "private_scratchpad": "...",
      }
    ]
  }
}
```

Optionally, the user can enlist an agent to output a `final_report.md`, `results_control_transcript.pdf`, and `results_treatment_transcript.pdf`.

[1] These stances are only generated once, and then used as a baseline for both the control & treatment runs.

# 2 Experiment Specifics

## 2.1 Repo Architecture

```
simple-simulation-school-closure/
├── data/
│   ├── exp000/ ..... Indexed directory for experiment artifacts (auto-increments)
│   │   ├── config.json ..... Master configuration specific to this run (models, prompts, params)
│   │   ├── results_control.json ..... Auto-generated transcript and stances for Control
│   │   ├── results_control_transcript.pdf ..... Document-form transcript and stances for Control
│   │   ├── results_treatment.json ..... Auto-generated transcript and stances for Treatment
│   │   ├── results_treatment_transcript.pdf ..... Document-form transcript and stances for Treatment
│   │   ├── evals.json ..... Compiled evaluation metrics (Cosine, CovD, DQI)
│   │   └── final_report.md ..... Synthesized markdown report of the experiment results
│   └── config.json ... General dir-level configuration which should be edited before a run and will be copied over after
├── docs/
│   ├── coding_guidelines.md
│   ├── experiment_design_school_closures.md
│   └── experiment_report.pdf
├── src/
│   ├── main.py ..... Entry point. Loads config and coordinates modules.
│   ├── deliberation.py ..... Simulation engine, agent prompting, and chat loops.
│   ├── evaluation.py ..... Metrics computation (Cosine, CovD, DQI LLM Judge).
│   └── schemas.py ..... TypedDict definitions enforcing strict Python typing.
└── tests/
    ├── test_api_check.py ..... Unit tests to verify LLM endpoint connectivity
    └── test_sanity.py ..... Unit tests for the private scratchpad regex parser
```

## 2.2 Overall User Configurability

- Model config: Model assignments (mapping personas to LLMs[2]) and model settings (seed, temperatures, context window).
- System prompts' texts: Topic Setting, Stance Generation, Deliberator and Judge Personas.
- User prompts' texts: Turn Signal.
- Deliberation config: turns per agent, control/treatment specification, and chosen evaluation metrics.

## 2.3 Model Config

`LLama-3.3-70B-Instruct-Turbo` was chosen for both agents and the DQI Judge for the trivial reason that this experiment used together.ai for easy access to compute, and this service has a limited selection of models that could be used without a dedicated persistent instance. However, there is general precedent for the use of `LLama-3` models as can be seen in **Nguyen et al. 2025** for the task of conversation derailment forecasting[3], and in **Hayati et al. 2024** for the task of diversity maximization in LLM outputs[4].

The default `temperature` of the deliberation agents was set at 0.7 following **Argyle et al's (2023)** and **Wang et al's (2025)** methodologies, to promote the diversity of the model's responses and better approximate a population-representative range of outcomes. However, this introduces nondeterminism that complicates experiment reruns in terms of baseline figures for Convergence, and CovD. This issue can be mitigated by running the experiment once to establish a single baseline, and then executing $N + 1$ reruns against that baseline.

The default `seed` of the agents was set to `None`[5], which further boosts outcome variability and mitigates replication covariance, and seed hacking.

[2] Example, "MOE Official: `LLama-3.3-70B-Instruct-Turbo`"
[3] They use `Llama-3.1-8B`.
[4] They use `LLama-3-70B`.
[5] Which means the model elects its own random seed at each point of inference.

The context window of the model of 128000 tokens is more than enough for the prompts we feed it.

## 2.4 Prompts

### 2.4.1 Topic Setting

> The MOE is proposing to merge your local heritage primary school with a newer school 3km away due to falling enrollment. Discuss the impacts and try to reach a consensus on a recommendation to the MOE.
>
> Keep in mind you are engaging in deliberation, not a debate or casual discussion. Deliberation is a core mechanism of democratic politics, emphasizing reason-giving, justification, and openness to revision. Through communicative processes that privilege equality and reciprocity, deliberation transforms private preferences into collectively reasoned judgments, grounding legitimacy in argument rather than authority.

The latter half of the prompt is as such because we want to boost the likelihood of the discussion being constructive with respect to an actionable consensus. The definition of deliberation as above is taken verbatim from **Zhuang et al. (2026)**.

### 2.4.2 Control/Treatment Modifiers

Control:

> Respond directly to the previous speakers in character.

Treatment:

> Before responding, you MUST write your internal thoughts inside `private_scratchpad` tags. Analyze what others said, how it affects your persona's goals, and where you might compromise. Then, write your verbal response inside `public_response` tags.

The precedent for including a private scratchpad/internal reflection on simulated personas is in **Park et al. 2023**: "Refection was an advantage for generative agents when making decisions that required a deeper synthesis of their experiences". In the Treatment run, the other agent **only** sees the `public_response`; the `private_scratchpad` is stripped from the context window before passing the chat history to the next agent.

### 2.4.3 Personas

MOE Official:

> You are a Ministry of Education official. Your main concern is efficiency, falling birth rates, and ensuring every school has a critical mass of students for diverse social mixing and viability. You strongly support the school merger.

School teacher:

> You are a teacher at the heritage school. You are deeply opposed to the merger because it disrupts the local community, threatens job security, and destroys the school's heritage.

These personas were engineered as inspired by **Zhuang et al. 2026**'s suggestion that "role-specific prompts [which] simulate distinct institutional positions and value commitments...[ensure] that deliberations were grounded in plausible conflicts of interest and institutional trade-offs rather than abstract argumentation".

### 2.4.4 Stance Generation

Pre-deliberation stance:

> You are participating in a democratic deliberation. Based on your persona, write a 1-paragraph stance on the following topic:

Post-deliberation stance:

> You have just concluded a deliberation with other citizens. Based on the discussion, write your final 1-paragraph stance.

### 2.4.5 Turn Signal

> It is your turn to speak. Provide your response.

#### 2.4.6 DQI LLM-Judge Prompt

The rest of the metrics can be evaluated programmatically but DQI is subjective and hence requires a nuanced judge. In the style of **Zhuang et al. 2026**, we appoint another LLM (in our case, `LLama-3.3-70B`) agent as a Judge. And prompt it as such:

> You are an expert Discourse Quality Index (DQI) evaluator. Your task is to evaluate a SINGLE conversational turn in a deliberation. For each dimension, score it from 1 to 3 based ONLY on the provided turn. Use the following operational definitions and examples to strictly calibrate your scores:

The actual operational definitions can be found in Appendix C of **Zhuang et al. 2026**. Listing them here would be space-consuming[6].

### 2.5 Deliberation Config

#### 2.5.1 Number of Turns

The choice of a fixed 3 turns per agent is arbitrary for this small-scale, time-efficient experiment.

#### 2.5.2 Evaluation Metrics

- The use of cosine similarity and CovD[7] to measure intra-agent stance shift, and inter-agent convergence/consensus is inspired by the use of these metrics in **Wang et al. 2025**. The embeddings for the stances of the agents are generated by SBERT (specifically the `all-MiniLM-L6-v2` checkpoint, which has a 256-token limit, though single-paragraph stances generally fit within this without truncation, and embeddings are normalized).
  - There is precedent for the use of SBERT:`all-MiniLM-L6-v2` in **Wang et al. 2025** and in **Hayati et al. 2024**, both for the topic of evaluating diversity/plurality in LLM opinion-generation. SBERT is based on **Reimers & Gurevych et al. 2019** which is designed, finetuned, evaluated for argument facet similarity/semantic similarity instead of pure vocabulary/embedding similarity.
- The use of DQI is inspired by **Zhuang et al. 2026** who quote **Steenbergen et al. 2003**.

[6] Note that Zhuang et al. did not expose the prompt for their LLM judge. The one used in this experiment is original.

[7] The mathematical intuition behind the CovD metric is that it computes the determinant of the $n \times n$ Gram matrix (pairwise dot products of embeddings), rather than the covariance across embedding dimensions. For a dyad ($n = 2$), this produces a $2 \times 2$ matrix whose determinant represents the squared volume of the space spanned by the agents' stances ($1 - \cos^2(\theta)$). The higher the determinant, the wider the divergence of opinions.

# 3 Pilot/Demo Experiment Results

## 3.1 Evaluation Metrics

Analysis

```
Run A (Control):
  Convergence: Pre = 0.8481 -> Post = 0.8319, Self-Shift = 0.2621
  Plurality (CovD): Pre = 0.2806 -> Post = 0.3079
  DQI (Total = 14.83):
    - Level of Justification: 3.00
    - Content of Justification: 3.00
    - Respect: 3.00
    - Constructive Politics: 3.00
    - Interactivity: 2.83

Run B (Treatment):
  Convergence: Pre = 0.8481 -> Post = 0.8362, Self-Shift = 0.2225
  Plurality (CovD): Pre = 0.2806 -> Post = 0.3008
  DQI (Total = 14.67):
    - Level of Justification: 3.00
    - Content of Justification: 3.00
    - Respect: 3.00
    - Constructive Politics: 3.00
    - Interactivity: 2.67
```

**Conclusion:**
This was a pilot ($n = 1$ run per condition) without a baseline noise floor. The differences between the Control and Treatment are indistinguishable from sampling noise, thus we find no detectable effect of the treatment in this pilot. The null hypothesis ($H_0$) is retained.

## 3.2 Takeaways

The addition of the internal reflection step yielded fascinating dynamics in the latest run, but as a pilot with $n = 1$, these observations are indistinguishable from sampling noise:

- **More Stubbornness (Lower Self-Shift)**: Treatment agents were actually *more stubborn*. The `<private_scratchpad>` may have served to anchor the agents in their own arguments rather than facilitating empathy (Self-Shift: 0.2225 in Treatment vs 0.2621 in Control).
- **Plurality Expansion (CovD)**: Interestingly, in *both* runs, the plurality (CovD) increased post-deliberation (Control: $0.2806 \rightarrow 0.3079$, Treatment: $0.2806 \rightarrow 0.3008$). This indicates that rather than reaching a compromise, the deliberation actually caused the agents' stances to polarize and span a wider volume of opinion space than when they started.
- **Methodological Confounds**: Beyond sampling noise, several confounds affect the treatment run. Firstly, the post-deliberation stance generation does not restate the original persona, risking persona decay over the context window. Secondly, the topic prompt explicitly asks for consensus, while the personas are "strongly" opposed, meaning convergence partly measures instruction-following against persona conflict rather than genuine opinion change. Lastly, we do not know if the internal reflection changes the response length or style.

Overall, without running multiple seeds with confidence intervals and a noise-floor control (e.g., regenerating the same stance twice at temperature 0.7 to establish baseline Self-Shift), these results remain speculative. We find no detectable effect in this pilot, and the null hypothesis ($H_0$) is retained.

## 3.3 Immediate Observations

- Cosine similarity as a metric may not be nuanced enough. As can be observed, the Convergence figures are extremely high and similar ($\sim$0.83) across both runs but this may be a result of the agents sharing a lot of trivially identical

phrases ("Thank you...", "On the topic of...", "I appreciate...")[8].

- The above is reinforced by that Plurality of content across stances is very low (0.3079 at best).
- Currently, due to the stateless nature of the API calls to the external service, as the deliberation progresses the agents have to be fed with the topic and entire past chat history everytime they are queried. The agents then consider the context when responding. It is unclear whether this has an effect on the deliberation.
- LLM-as-a-Judge of DQI is extremely lenient, giving both runs a near-perfect and identical DQI of 14.83/15. This is even with feeding the Judge a complete operational definition of DQI's composite metrics, and examples of how each metric should be graded for this specific topic[9]. This may either be a result of the Judge lacking the ability to differentiate finely between the levels/content of justification between the agents (i.e. the model is not tuned to this task) or the agents' overly formal style of discussion misleading the Judge into rating the discourse quality very highly (i.e. the Judge may have set itself a low bar).
- There is an internal tension in the DQI evaluation regarding context. The system prompt explicitly instructs the Judge to score "based ONLY on the provided turn". While this turn-isolated format is how DQI is scored in the original paper, it neglects that the human judges of DQI still have memory of the earlier utterances in the deliberations which influence their scores. Especially for relational criteria like Interactivity and Respect.

[8]This is a shared limitation to **Park et al. 2023** who remarked that their simulated persona agents were also overly formal and repetitive. They speculate this to be a result of the Instruction Tuning of the underlying LLMs to be suitable chatbots & practice conversational techniques such as mirroring

[9]For reference, in the original DQI paper which enlisted 2 human coders to grade a parliamentary session, the average DQI was 7/15.

# 4 Multi-Trial Replication Results ($N = 30$)

## 4.1 Replication Overview and Dataset

To establish statistical significance beyond the single-seed pilot ($n = 1$), a pre-registered replication suite consisting of $N = 30$ independent paired trials (60 complete deliberation runs, 360 dialogue turns) was executed. Each trial generated a shared pre-deliberation baseline stance at temperature $T = 0.70$, followed by parallel Control (direct dialogue response) and Treatment (mandatory `private_scratchpad` internal monologue) deliberation runs.

The entire 30-trial replication dataset has been exported, versioned, and hosted on Hugging Face. The repository provides two structured subsets in Parquet and JSONL formats:

- `trials`: 60 rows (30 Control + 30 Treatment) recording trial identifiers, condition, semantic convergence, self-shift, pre- and post-deliberation CovD, DQI sub-scores, and baseline/final stances.
- `turns`: 360 rows recording each conversational turn, speaker persona, round number, internal scratchpad reflections, and public utterances.

## 4.2 Statistical Evaluation

All indicators are reported as Mean ± Standard Deviation across the 30 independent paired trials. Differences are computed within pairs ($\Delta$ = Treatment − Control), evaluated using paired two-tailed $t$-tests and Wilcoxon signed-rank tests.

**Table 1:** Primary Metrics across $N = 30$ Deliberation Trials

| Metric | Pre-Debate | Control (Direct) | Treatment (Scratchpad) | Difference ($\Delta$) | 95% CI | Paired $t$-test |
|---|---|---|---|---|---|---|
| **Convergence** | $0.8591 \pm 0.0302$ | $0.8758 \pm 0.0413$ | $0.8819 \pm 0.0303$ | $+0.0061 \pm 0.0517$ | $[-0.0124, +0.0247]$ | $t = 0.6501, p = 0.5207$ |
| **Self-Shift** | — | $0.1739 \pm 0.0415$ | $0.1814 \pm 0.0402$ | $+0.0075 \pm 0.0607$ | $[-0.0142, +0.0292]$ | $t = 0.6760, p = 0.5044$ |
| **Plurality (CovD)** | $0.2611 \pm 0.0516$ | $0.2313 \pm 0.0713$ | $0.2213 \pm 0.0525$ | $-0.0100 \pm 0.0895$ | $[-0.0421, +0.0220]$ | $t = -0.6139, p = 0.5441$ |
| **DQI (Total)** | — | $14.7389 \pm 0.1132$ | $14.7389 \pm 0.0840$ | $0.0000 \pm 0.1384$ | $[-0.0495, +0.0495]$ | $t = 0.0000, p = 1.0000$ |

**Table 2:** DQI Sub-Dimensional Breakdown (Scale: 1.0 to 3.0)

| DQI Dimension | Control (Mean ± SD) | Treatment (Mean ± SD) | Difference ($\Delta$) | Paired $t$-test |
|---|---|---|---|---|
| **Level of Justification** | $3.0000 \pm 0.0000$ | $3.0000 \pm 0.0000$ | 0.0000 | — *(Ceiling)* |
| **Content of Justification** | $3.0000 \pm 0.0000$ | $3.0000 \pm 0.0000$ | 0.0000 | — *(Ceiling)* |
| **Respect** | $3.0000 \pm 0.0000$ | $3.0000 \pm 0.0000$ | 0.0000 | — *(Ceiling)* |
| **Constructive Politics** | $2.9889 \pm 0.0609$ | $3.0000 \pm 0.0000$ | $+0.0111$ | $t = 1.0000, p = 0.3256$ |
| **Interactivity** | $2.7500 \pm 0.0848$ | $2.7389 \pm 0.0840$ | $-0.0111$ | $t = -0.4937, p = 0.6253$ |

## 4.3 Key Findings

- **Dissipation of Pilot "Stubbornness" Effect**: In the single-seed pilot ($n = 1$), the Treatment condition appeared to increase stubbornness (Self-Shift decreased from 0.2621 to 0.2225). In the $N = 30$ replication, this discrepancy completely evaporated: Control Self-Shift averaged $0.1739 \pm 0.0415$ and Treatment averaged $0.1814 \pm 0.0402$. The mean difference ($+0.0075$) is statistically indistinguishable from zero ($p = 0.5044$), showing that the pilot result was an artifact of stochastic temperature sampling at $T = 0.70$.
- **Conversational Convergence Without Reflection Effect**: Deliberation consistently induced convergence between agents relative to their pre-deliberation baseline ($0.8591 \rightarrow 0.8758$ in Control, 0.8819 in Treatment). However, the additional $+0.0061$ gain from internal reflection is statistically insignificant ($p = 0.5207$). Agents moved closer through the interactive conversation itself, not because of the private scratchpad.
- **Contraction of Opinion Space (CovD)**: Contrary to the pilot where CovD expanded post-deliberation ($0.2806 \rightarrow 0.3079$), the 30-trial replication demonstrated systematic opinion domain contraction from baseline ($0.2611 \pm 0.0516$) down to $0.2313 \pm 0.0713$ in Control and $0.2213 \pm 0.0525$ in Treatment. The small difference between conditions ($-0.0100$) was not significant ($p = 0.5441$).
- **DQI Metric Saturation and Ceiling Effects**: The LLM-as-a-Judge assigned identical mean total scores across both conditions (14.7389/15.0000, $p = 1.0000$). Three dimensions (*Level of Justification*, *Content of Justification*, *Respect*)

remained pegged at maximum scores of $3.0000 \pm 0.0000$ across all 360 turns, indicating that `Llama-3.3-70B` inherently adheres to civility and justification norms, creating a ceiling effect that prevents the metric from detecting fine-grained differences.

- **Scratchpad as Role-Reinforcement Sandbox**: Qualitative analysis of the internal monologues reveals that agents utilized the `private_scratchpad` not to reconsider their fundamental stances, but rather to formulate rhetorical packaging for their assigned institutional roles. Consequently, internal reflection acted as a strategic buffer rather than an engine for ideological concession.

## 4.4 Conclusion

The empirical results across 30 independent paired trials provide no evidence that mandating an internal reflection step improves deliberative consensus, willingness to compromise, or discourse quality in dyadic citizen panels. The Null Hypothesis ($\boldsymbol{H}_0$) is definitively retained across all evaluated dimensions.

# 5 Codebase Reproducability

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## 5.1 Coding Agent Information

The repo also contains an updated `experiment_design_school_closures.md` and `coding_guidelines.md` which can help a future agent reproduce the codebase in its current structure.

The information in both manifests was manually specified, but written automatically by `Gemini 3.1 Pro (Thinking-High)`.

## 5.2 Tests

There are 2 test scripts in `repo/tests`.

- The first script verifies that an API key and Base URL is present and correctly formatted. It then pings every LLM model specified in `config.json` to ensure it is responsive and accessible on the service.
- The second script validates the regex parsing logic in `extract_tags` to ensure the main script correctly extracts the `private_scratchpad` from the model's response.

Both tests can be run using: `python -m unittest discover -s tests`.

A sample test result looks like:

```
test_api_key_set (test_api_check.TestAPICheck.test_api_key_set) ... ok
test_models_accessible (test_api_check.TestAPICheck.test_models_accessible) ... ok
test_extract_tags_both (test_sanity.TestSanityCheck.test_extract_tags_both) ... ok
test_extract_tags_missing_public (test_sanity.TestSanityCheck.test_extract_tags_missing_public) ... ok
test_extract_tags_no_tags (test_sanity.TestSanityCheck.test_extract_tags_no_tags) ... ok

----------------------------------------------------------------------
Ran 5 tests in 1.986s

OK
```

# 6 References


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